

More algorithms

Explicit implementations of some statistic formulas

Pearson's linear correlation

- Measures the strength and direction of the linear relationship between two numeric variables
- r ranges from -1 to 1
- $r = 1$: perfect positive linear relationship
- $r = -1$: perfect negative linear relationship
- $r = 0$: no linear relationship

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

Pearson's linear correlation

- R makes available an efficient implementation
- As a programming exercise we will see three implementations
 - With full loops
 - With partial optimization using the `mean()` function
 - With full vectorisation

```
cor(x, y = NULL, use = "everything",  
    method = c("pearson", "kendall", "spearman"))
```

Pearson's linear correlation - test

1. Generate synthetic test data
2. Import a version of the function
3. compute the correlation
4. show the plot of the data and the computed correlation
5. compare the computed correlation with the result of the native R function

Pearson's linear correlation

`gen-corr.R`

- Generates two numeric vectors x and y of length n with Pearson correlation approximately equal to a chosen value $-1 \leq c \leq 1$
- The construction is based on the formula: $y = cz_1 + \sqrt{1 - c^2}z_2$
 - z_1 is a standardized random vector
 - z_2 is another standardized random vector made orthogonal to z_1
 - c controls the desired correlation between x and y
 - $\sqrt{1 - c^2}$ controls the remaining independent variation
- Because z_1 and z_2 are orthogonal and standardized, the resulting vectors have correlation exactly equal to c , up to numerical precision.
- **Local variables**
 - x : first generated vector
 - z : auxiliary random vector
 - $x_centered$: centered version of x
 - $z_centered$: centered version of z
 - $projection$: projection of $z_centered$ onto $x_centered$
 - $z_orthogonal$: component of $z_centered$ orthogonal to $x_centered$
 - $x_standard$: standardized version of x